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Wheel for closing system

Abstract

A wheel adapted for use with a trench closing assembly. The wheel has a circumferential tread coaxial with a central axis of the wheel, the circumferential tread extending axially an axial width. A continuous wall member joined with the circumferential tread extends radially inward to a central opening and axially outward from an inner edge of the circumferential tread by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width, whereby the circumferential tread and the wall member together defining a shape in the form of an uppercase Greek letter sigma.

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Background/Summary

BACKGROUND

(1) It is well known that good seed-to-soil contact within the seed trench is a critical factor in uniform seed emergence and high yields. To obtain good seed-to-soil contact, once seeds are planted in a seed trench, the seed trench needs to be closed so that soil surrounds the seed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side elevation view of an embodiment of a row unit of an agricultural planter.
- (2) FIG. 2 is a diagram of a system for implementing operational control of the closing wheel assembly based on signals generated by the trench closing sensor.
- (3) FIG. 3 is a side elevation view of an embodiment of a trench closing assembly.
- (4) FIG. 4 is a top plan view of the trench closing assembly of FIG. 3.
- (5) FIG. 5 is a bottom plan view of the trench closing assembly of FIG. 3.
- (6) FIG. 6 is a perspective view of the trench closing assembly of FIG. 3.
- (7) FIG. 7 is a perspective view of the frame of the trench closing assembly of FIG. 3.
- (8) FIG. 8 is a bottom plan view of the frame of the trench closing assembly of FIG. 7.
- (9) FIG. 9 is a side elevation view of the frame of the trench closing assembly of FIG. 7.
- (10) FIG. 10 is a perspective view of the frame of the trench closing assembly of FIG. 7 with the actuator removed.
- (11) FIG. 11 is a bottom perspective view of the frame of the trench closing assembly of FIG. 10 with a swing arm removed.
- (12) FIG. 12A is top perspective view of the main frame of the trench closing assembly.
- (13) FIG. 12B is bottom perspective view of the main frame of the trench closing assembly.
- (14) FIG. 13 is a perspective view of the press wheel frame of the trench closing assembly.
- (15) FIG. 14 is perspective view of the transfer bar of the trench closing assembly.
- (16) FIG. 15 is a perspective view of the transfer bar and swing arms of the trench closing assembly.
- (17) FIG. 16 is a perspective view of the transfer bar and swing arms of FIG. 15 and further including the press wheel frame.
- (18) FIG. 17 is a side elevation view of a portion of the trench closing assembly.
- (19) FIG. 18 is a perspective view of portions of the trench closing assembly.
- (20) FIG. 19 is a perspective view of portions of the trench closing assembly.
- (21) FIG. 20 is a perspective view of portions of the trench closing assembly.
- (22) FIG. 21 is a side elevation view of a wheel according to one embodiment.
- (23) FIG. 22 is a perspective view of the wheel of FIG. 21.
- (24) FIG. 23 is a side elevation view of an embodiment of the spoke disk of the wheel of FIG. 21.
- (25) FIG. 24 is a perspective view of the spoke disk of FIG. 23 showing some of the flanges received within the notches of respective spoke arms.
- (26) FIG. 25 is a schematic illustration of a fluid system according to one embodiment.
- (27) FIG. 26 is a schematic illustration of a fluid system according to another embodiment.
- (28) FIG. 27 is a perspective view of the trench closing assembly with a force sensor.
- (29) FIG. 28 is a side elevation view of the modified mounting arm with a position sensor.
- (30) FIG. 29 is a perspective view of an embodiment of a trench closing assembly.
- (31) FIG. 30 is a top plan view of the trench closing assembly of FIG. 29.
- (32) FIG. 31 is a bottom plan view of the trench closing assembly of FIG. 29.
- (33) FIG. 32 is a side elevation view of the trench closing assembly of FIG. 29.

(34) FIG. 33 is a perspective view of the trench closing assembly of FIG. 29 with the actuator and a wheel removed.
(35) FIG. 34 is a perspective view of the trench closing assembly of FIG. 33 with the actuator base removed.
(36) FIG. 35 is a side elevation view of the trench closing assembly of FIG. 34 with the wheels removed.
(37) FIG. 36 is a bottom perspective view of the trench closing assembly of FIG. 35.
(38) FIG. 37 is a perspective view of the trench closing assembly of FIG. 35 with additional parts removed.
(39) FIG. 38A is a side elevation view of the mounting arm.
(40) FIG. 38B is a perspective view of the press wheel frame.
(41) FIG. 39 is a perspective view of the trench closing assembly of FIG. 37 with additional parts removed.
(42) FIG. 40 is a bottom perspective view of the trench closing assembly of FIG. 39.
(43) FIG. 41 is a perspective view of the trench closing assembly of FIG. 39 with additional parts removed.
(44) FIG. 42 is a bottom perspective view of the crossbar connected to the arms.
(45) FIG. 43 is a side elevation view of FIG. 42.
(46) FIG. 44 is a perspective view of a press wheel according to one embodiment.
(47) FIG. 45 is a side elevation view of the press wheel of FIG. 44.
(48) FIG. 46 is a side elevation view of the press wheel of FIG. 45 with the hub removed.
(49) FIG. 47 is a side elevation view of a trench closing assembly with adjustable depth according to one embodiment.
(50) FIG. 48 is a top plan view of the trench closing assembly of FIG. 47.
(51) FIG. 49 is a schematic of a valve system.
(52) FIG. 50 is a perspective view of an embodiment of a trench closing assembly.
(53) FIG. 51 is a top plan view of the trench closing assembly of FIG. 50.
(54) FIG. 52 is a bottom plan view of the trench closing assembly of FIG. 50.
(55) FIG. 53 is a side elevation view of the trench closing assembly of FIG. 50.
(56) FIG. 54 is a perspective view of the trench closing assembly of FIG. 50 with a wheel removed.
(57) FIG. 55 is a perspective view of the trench closing assembly of FIG. 50 with the wheels, scrapper, and actuator removed.
(58) FIG. 56 is a side elevation view of the trench closing assembly of FIG. 55.
(59) FIG. 57 is a perspective view of the trench closing assembly of FIG. 55 with an arm removed.
(60) FIG. 58 is a perspective view of the trench closing assembly of FIG. 57 with the connection bracket removed.
(61) FIG. 59 is a perspective view of a portion of the trench closing assembly of FIG. 50.
(62) FIG. 60 is a perspective view of the trench closing assembly of FIG. 59 with the knob and force sensor removed.
(63) FIG. 61 is a perspective view of the trench closing assembly of FIG. 50 with parts removed for viewing.
(64) FIG. 62 is a side elevation view of the trench closing assembly of FIG. 61.
(65) FIG. 63 is a perspective view of a portion of the trench closing assembly of FIG. 50.
(66) FIG. 64 is a perspective view of an arm.
(67) FIG. 65 is a perspective view of the connection bracket.
(68) FIG. 66 is a perspective view of a packing wheel system with a force sensor of one embodiment.
(69) FIG. 67 is a perspective view of a packing wheel system with a force sensor of one embodiment.
(70) FIG. 68 is a perspective view of a packing wheel system with a force sensor of one embodiment.
(71) FIG. 69 is a perspective view of a packing wheel system with a force sensor of one embodiment.
(72) FIG. 70 is a perspective view of a trench closing assembly further including a soil leveler according to one embodiment.
(73) FIG. 71 is a side elevation view of the trench closing assembly of FIG. 70.
(74) FIG. 72 is a side elevation view of the soil leveler of FIG. 70.
(75) FIG. 73 is a perspective view of the soil leveler of FIG. 72.
(76) FIG. 74 is a bottom plan view of the soil leveler of FIG. 72.
(77) FIG. 75 is a perspective view of a trench closing assembly further including a soil leveler according to one embodiment.
(78) FIG. 76 is a side elevation view of the trench closing assembly of FIG. 75.
(79) FIG. 77 is a perspective view of the soil leveler of FIG. 75.
(80) FIG. 78 is a side elevation view of the soil leveler of FIG. 77.
(81) FIG. 79 is a perspective view of a portion of the soil leveler of FIG. 75.
(82) FIG. 80 is a left side elevation view of an embodiment of a trench closing assembly.
(83) FIG. 81 is a top plan view of the trench closing assembly of FIG. 80.
(84) FIG. 82 is a bottom plan view of the trench closing assembly of FIG. 80.
(85) FIG. 83 is a top perspective view of the trench closing assembly of FIG. 80 with some parts removed.
(86) FIG. 84 is a bottom perspective view of the trench closing assembly of FIG. 83.
(87) FIG. 85 is a bottom perspective view of the trench closing assembly of FIG. 84 with some parts removed.
(88) FIG. 86 is a bottom perspective view of the trench closing assembly of FIG. 85 with some parts removed.
(89) FIG. 87 is a top perspective view of the trench closing assembly of FIG. 86.
(90) FIG. 88 is a top perspective view of the frame of the trench closing assembly of FIG. 80.
(91) FIG. 89 is a bottom perspective view of the frame of FIG. 88.
(92) FIG. 90 is bottom plan view of the frame of FIG. 88.
(93) FIG. 91 is top perspective view of the frame of the trench closing assembly of FIG. 80.
(94) FIG. 92 is a bottom perspective view of the frame of FIG. 91.
(95) FIG. 93 is a top perspective view of the connection bracket of the frame of FIG. 91.
(96) FIG. 94 is a bottom perspective view of the connection bracket of FIG. 93.
(97) FIG. 95 is a top perspective view of the attachment bracket of the frame of FIG. 91.
(98) FIG. 96 is a bottom perspective view of the attachment bracket of FIG. 95.
(99) FIG. 97 is top perspective view of the mounting arm of the trench closing assembly of FIG. 80.
(100) FIG. 98 is a bottom perspective view of the mounting arm of FIG. 97.
(101) FIG. 98 is a front elevation view of the handle assembly of the trench closing assembly of FIG. 80.
(102) FIG. 99 is a left side elevation view of the handle assembly of FIG. 98.
(103) FIG. 101 is a perspective view of the shank of the handle assembly of FIG. 98.
(104) FIG. 102 is a bottom perspective view of the pressure sensor of the handle assembly of FIG. 98.
(105) FIG. 103 is a top perspective view of the pressure sensor of FIG. 102.
(106) FIG. 104 is a perspective view of the handle assembly and mounting arm of the trench closing assembly of FIG. 80.
(107) FIG. 105 is a side elevation view of a press wheel according to one embodiment.
(108) FIG. 106 is a perspective view of the press wheel of FIG. 105.

- (109) FIG. 1 is a front elevation view of the press wheel of FIG. 105.
- (110) FIG. 108 is a perspective view of an embodiment of a fluid control assembly.
- (111) FIG. 109 is an exploded perspective view of the fluid control assembly of FIG. 108.
- (112) FIG. 110 is a top plan view of the fluid control assembly of FIG. 108.
- (113) FIG. 111 is a perspective view of the board of the fluid control assembly of FIG. 108.
- (114) FIG. 112 is a bottom perspective view of the housing of the fluid control assembly of FIG. 108.
- (115) FIG. 113 is a section elevation view of the housing of FIG. 112.
- (116) FIG. 114 is a bottom perspective view of the section of the housing of FIG. 113.
- (117) FIG. 115 is an enlarged perspective view of one embodiment of the closing wheel shown in FIG. 80.
- (118) FIG. 116 is a side elevation of the closing wheel of FIG. 115.
- (119) FIG. 117 is a perspective view of a trench closing assembly with an embodiment of a wheel which may be used as press wheels.
- (120) FIG. 118 is a perspective view of one of the wheels of FIG. 117 shown in isolation.
- (121) FIG. 119 is a front elevation view of the wheel of FIG. 118.
- (122) FIG. 120 is a rear elevation view of the wheel of FIG. 118.
- (123) FIG. 121 is another perspective view of the wheel of FIG. 118 with the hub removed.
- (124) FIG. 122 is a side elevation view of the wheel of FIG. 118.
- (125) FIG. 123 is a cross-section view of the wheel of FIG. 118.
- (126) FIG. 124 is a perspective view of the cross-sectional view of FIG. 123.
- (127) FIG. 125 is a perspective view of an outer side of another embodiment of a wheel which may be used as one of the press wheels for a trench closing assembly.
- (128) FIG. 126 is a perspective view of the inner side of the wheel of FIG. 125.
- (129) FIG. 127 is an elevation view of the outer side of the wheel of FIG. 125.
- (130) FIG. 128 is an elevation view of the inner side of the wheel of FIG. 125.
- (131) FIG. 129 is a side elevation view of the wheel of FIG. 125.
- (132) FIG. 130 is a cross-section view of the wheel of FIG. 125.

DESCRIPTION

(133) Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIG. 1 illustrates an embodiment of an agricultural planter row unit 200. The row unit 200 is comprised of a frame 204 pivotally connected to a toolbar 202 by a parallel linkage 206 enabling each row unit 200 to move vertically independently of the toolbar 202. The frame 204 operably supports one or more hoppers 208, a seed meter 210, a seed delivery mechanism 212, a downforce control system 214, a seed trench opening assembly 220, a trench closing assembly 250, an optional packer wheel assembly 260, and an optional row cleaner assembly 270. It should be understood that the row unit 200 shown in FIG. 1 may be for a conventional planter or the row unit 200 may be a central fill planter, in which case the hoppers 208 may be replaced with one or more mini-hoppers and the frame 204 modified accordingly as would be recognized by those of skill in the art.

(134) The optional downforce control system 214 is disposed to apply lift and/or downforce on the row unit 200 such as disclosed in U.S. Publication No. US2014/0090585, which is incorporated herein in its entirety by reference. The downforce applied by downforce control system 214 can be determined by methods disclosed in U.S. Pat. No. 9,173,339, which is incorporated herein in its entirety by reference.

(135) The seed trench opening assembly 220 includes a pair of opening discs 222 rotatably supported by a downwardly extending shank member 205 of the frame 204. The opening discs 222 are arranged to diverge outwardly and rearwardly so as to open a v-shaped trench 10 in the soil 11 as the planter traverses the field. The seed delivery mechanism 212, such as a seed tube or seed conveyor, is positioned between the opening discs 222 to deliver seed from the seed meter 210 into the opened seed trench 10. The depth of the seed trench 10 is controlled by a pair of gauge wheels 224 positioned adjacent to the opening discs 222. The gauge wheels 224 are rotatably supported by gauge wheel arms 226 which are pivotally secured at one end to the frame 204 about pivot pin 228. A rocker arm 230 is pivotally supported on the frame 204 by a pivot pin 232. It should be appreciated that rotation of the rocker arm 230 about the pivot pin 232 sets the depth of the trench 10 by limiting the upward travel of the gauge wheel arms 226 (and thus the gauge wheels) relative to the opening discs 222. The rocker arm 230 may be adjustably positioned via a linear actuator 234 mounted to the row unit frame 204 and pivotally coupled to an upper end of the rocker arm 230. The linear actuator 234 may be controlled remotely or automatically actuated as disclosed, for example, in International Publication No. WO2014/186810, which is incorporated herein in its entirety by reference.

(136) An optional downforce sensor 238 is configured to generate a signal related to the amount of force imposed by the gauge wheels 224 on the soil. In some embodiments the pivot pin 232 for the rocker arm 230 may comprise the downforce sensor 238, such as the instrumented pins disclosed in U.S. Pat. No. 8,561,472, which is incorporated herein in its entirety by reference.

(137) An optional seed meter 210 may be any commercially available seed meter, such as a finger-type meter or a vacuum seed meter. One exemplary vacuum seed meter is the VSet® meter, available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568.

(138) An optional packer wheel assembly 260 comprises an arm 262 pivotally attached to the row unit frame 204 and extends rearward of the closing wheel assembly 250 and in alignment therewith. The arm 262 rotatably supports a packer wheel 264. An actuator 266 is pivotally attached at one end to the arm 262 and at its other end to the row unit frame 204 to vary the amount of downforce exerted by the packer wheel 264 to pack the soil over the seed trench 10.

(139) An optional row cleaner assembly 270 may be the CleanSweep® system available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568. The row cleaner assembly 270 includes an arm 272 pivotally attached to the forward end of the row unit frame 204 and aligned with the trench opening assembly 220. A pair of row cleaner wheels 274 are rotatably attached to the forward end of the arm 272. An actuator 276 is pivotally attached at one end to the arm 272 and at its other end to the row unit frame 204 to adjust the downforce on the arm to vary the aggressiveness of the action of the row cleaning wheels 274 depending on the amount of crop residue and soil conditions.

(140) Referring to FIG. 2, a monitor 300 is visible to an operator within the cab of a tractor pulling the planter. The monitor 300 may be in signal communication with a GPS unit 310, the trench closing assembly actuator 256 and the optional packer wheel assembly actuator 266 to enable operational control of the trench closing assembly 250 and the optional packer wheel assembly 260 based on the signals generated by trench closing sensors 1000, which are described in International Publication No. WO2017/197274, which is incorporated herein by reference in its entirety. Also as discussed later, the monitor 300 may be programmed to display operational recommendations based on the signals generated by the trench closing sensors 1000. The monitor 300 may also be in signal communication with the row cleaner actuator 276, the downforce control system 214, the depth adjustment actuator 234 to enable operational control of row cleaner assembly 270, the downforce control system 214 and the trench opening assembly 230, respectively.

(141) Trench Closing Assembly

(142) FIGS. 3 to 24 illustrate a trench closing assembly 250 according to one embodiment. The trench closing assembly 250 is adapted to be attached to row unit 200. Trench closing assembly 250 has a frame 251, an actuator 259, a pair of closing wheels 254, and optionally, a pair of press wheels 255. While illustrated with a pair of press wheels 255, a single press wheel 255 can be used. Actuator 259 can apply one force to frame 251, and this force can be divided between the closing wheels 254 and the press wheels 255.

(143) Actuator 259 can be any actuator that can apply a force. Examples of actuators include, but are not limited to, pneumatic actuators, hydraulic

actuators, electro-mechanical actuators, and electro-hydraulic actuators. In one embodiment, actuator 259 is a pneumatic actuator, such as an air bag or an air cylinder. A gas supply (not shown) can be connected to nozzle 1259 to supply gas (such as air) to actuator 259.

(144) Referring to FIGS. 7 and 12A and 12B, frame 251 has a main frame 1200 having a connection bracket 1201 that is adapted to be connected to row unit 200. Optionally, one or more bolts 1209 extending through apertures 1207 (FIG. 12A) in the connection bracket 1201 for mounting the frame 251 to the row unit 200. Extending up from connection bracket 1201 is actuator bracket 1210. Actuator bracket 1210 in one embodiment has a first bracket arm 1211-1, a second bracket arm 1211-2, and a cross connector 1212 connecting first bracket arm 1211-1 and second bracket arm 1211-2. Main frame 1200 has a first wing 1202-1 and a second wing 1202-2 extending laterally outward transverse to a direction of travel of trench closing assembly 250. As best viewed in FIGS. 12A-12B, extending downwardly from connection bracket 1201 is a first inner bracket 1204-1 and a second inner bracket 1204-2 to which the first wing 1202-1 and second wing 1202-2 are respectively attached. Extending downwardly from wings 1202-1 and 1202-2 at an end opposite to where the wings 1202-1, 1202-2 attach to inner brackets 1204-1, 1204-2 are outer brackets 1205-1 and 1205-2, respectively. Extending downwardly from wings 1202-1 and 1202-2 and between inner brackets 1204-1, 1204-2 and outer brackets 1205-1, 1205-2 are middle brackets 1203-1 and 1203-2, respectively.

(145) As seen in FIG. 18, an actuator base 1213 is connected to first base arm 1214-1 and 1214-2. A first transfer arm 1271-1 is connected to first base arm 1214-1, and a second transfer arm 1271-2 is connected to second base arm 1214-2. Connected to first transfer arm 1271-1 and second transfer arm 1271-2 at an end opposite to where the first base arm 1214-1 and second base arm 1214-2 connect is a transfer block 1269 through which a hole 1268 extends. As shown in FIG. 11, a transfer bar 1260 is disposed through the hole 1268 in the transfer block 1269.

(146) As seen in FIG. 14, transfer bar 1260 has a longitudinal bar 1262 and a transverse bar 1261. Transfer bar 1260 can be a unitary part, or (as shown) longitudinal bar is disposed through a bore in the transverse bar 1261. Longitudinal bar 1262 is generally oriented parallel to the direction of travel of trench closing assembly 250, and transverse bar 1261 is generally oriented transverse to the direction of travel. At each end of transverse bar 1261 are tabs 1263-1 and 1263-2. Tabs 1263-1 and 1263-2 are for mating with respective swing arms 1220-1 and 1220-2, via respective holes 1225-1 and 1225-2, as best seen in FIG. 15.

(147) Transfer bar 1260 divides the force from actuator 259 between closing wheels 254 (254-1, 254-2) and press wheels 255 (255-1, 255-2). Closing wheels 254-1 and 254-2 are connected to swing arms 1220-1 and 1220-2, respectively. Force applied to transfer bar 1260 is transferred through transverse bar 1261 to tabs 1263-1 and 1263-2. Also, force is transferred to press wheel frame 1250 via longitudinal bar 1262.

(148) As seen in FIG. 13, press wheel frame 1250 has a first arm 1251-1, a second arm 1251-2, a first cross brace 1253, which can optionally have a hole 1255 for receiving longitudinal bar 1262, a second cross brace 1252 having a hole 1254 for receiving longitudinal bar 1262, and a mounting arm 1256 to which press wheels 255-1 and 255-2 are mounted through bracket 1258. Press wheel frame 1250 is pivotably disposed between inner brackets 1204 (1204-1, 1204-2) and outer brackets 1205 (1205-1, 1205-2) about pivots 1206-1 and 1206-2, respectively.

(149) Swing arms 1220 (1220-1, and 1220-2) are pivotably disposed between inner brackets 1204 (1204-1, 1204-2) and outer brackets 1205 (1205-1, 1205-2) about pivots 1206-1 and 1206-2, respectively.

(150) As illustrated in FIGS. 17 to 19, first base arm 1214-1 and second base arm 1214-2 are pivotably connected to main frame 1200 through a first pivot arm 1280 and a second pivot arm 1285. First pivot arm 1280 has a first arm 1281-1 and a second arm 1281-2. First pivot arm 1280 and second pivot arm 1285 are pivotably disposed between inner brackets 1204-1 and 1204-2. First pivot arm 1280 pivots about pivot 1283, and second pivot arm pivots about pivot 1286.

(151) As illustrated in FIGS. 6 and 7, press wheels 255-1 and 255-2 are disposed on bracket 1258. Bracket 1258 has a plurality of holes 1257 for adjustable mating to mounting arm 1256. Wheels of different diameters can be attached to bracket 1258 or a different placement of wheels can be used to change the distribution of force with the adjustable mating.

(152) Optionally, a scrapper 1221 (1221-1) can be included. Scrapper 1221 is attached to swing arm 1220 and is disposed to receive a closing wheel 254. While shown with one scrapper 1221-1, a scrapper 1221-2 (not shown) can be attached to swing arm 1220-2 similar to scrapper 1221-1 being connected to swing arm 1220-1.

(153) FIGS. 29-43 illustrate another trench closing assembly 250A according to another embodiment. The trench closing assembly 250A is adapted to be attached to row unit 200. Trench closing assembly 250A has a frame 251A, an actuator 259, a pair of closing wheels 254-1, 254-2, and optionally, a press wheel 255A. As illustrated, the press wheel 255A may comprise a pair of press wheels 255A-1, 255A-2, but a single press wheel (not shown) may be utilized. Actuator 259 can apply one force to frame 251A, and this force can be divided between the closing wheels 254 and the press wheel 255A.

(154) Referring to FIG. 29, frame 251A has a main frame 1200A having a connection bracket 1201A that is adapted to be connected to row unit 200. As in the previous embodiment, one or more bolts 1209 (FIG. 33) can extend through apertures 1207 in connection bracket 1201A for mounting the frame 251A to the row unit 200. Extending up from connection bracket 1201A is actuator bracket 1210A. In one embodiment, actuator bracket 1210A has a first bracket arm 1211A-1, a second bracket arm 1211A-2, and a cross connector 1212A connecting first bracket arm 1211A-1 and second bracket arm 1211A-2. Actuator bracket 1210A can be made from separate parts or as a single part. As best viewed in FIGS. 34 and 37, extending downwardly from connection bracket 1201A are first bracket 1204A-1 and second bracket 1204A-2 with cross-braces 1208A-1 and 1208A-2 extending first bracket 1204A-1 and second bracket 1204A-2. Extending outward, transverse to the direction of travel, are optional stops 1229A. Stop 1229A-1 is disposed on first bracket 1204A-1, and stop 1229A-2 is disposed on second bracket 1204A-2. Stop 1229A-1 cooperates with stops 1228A (1228A-1a and 1228A-1b) on first swing arm 1220A-1 and stops 1259A (1259A-1a and 1259A-1b) on first arm 1251A-1. Stop 1229A-2 cooperates with stops 1228A (1228A-2a and 1228A-2b) on second swing arm 1220A-2 and stops 1259A (1259A-2a and 1259A-2b) on second arm 1251A-2. The angle of rotation of swing arms 1220A and first and second arms 1251A can be limited.

(155) As viewed in FIGS. 39 and 40, extending through first bracket 1204A-1 is pivot 1206A-1 transverse to the direction of travel. Extending through second bracket 1204A-2 is pivot 1206A-2 transverse to the direction of travel. Pivots 1206A-1 and 1206A-2 allow for pivoting of swing arms 1220A-1 and 1220A-2, respectively, first arm 1251A-1 and second arm 1251A-2, respectively, and arms 1295A-1 and 1295A-2, respectively.

(156) Swing arms 1220A-1 and 1220A-2 are pivotably disposed about pivots 1206A-1 and 1206A-2, respectively. Swing arms 1220A-1 and 1220A-2 are adjustable transverse to the direction of travel along pivot 1206A-1 and pivot 1206A-2, respectively. This allows for changing the width of spacing of closing wheels 254.

(157) Illustrated in FIG. 38A, press wheel frame 1250A has a first arm 1251A-1 and a second arm 1251A-2. First arm 1251A-1 and second arm 1251A-2 pivot about pivots 1206A-1 and 1206A-2, respectively. Mounting arm 1256A is connected to both first arm 1251A-1 and second arm 1251A-2. First arm 1251A-1 can optionally include stops 1259A-1a and 1259A-1b to limit the rotation of first arm 1251A-1 about pivot 1206A-1. Second arm 1251A-2 can optionally include stops 1259A-2a and 1259A-2b to limit the rotation of second arm 1251A-2 about pivot 1206A-2 at stops 1229A-1 and 1229A-2, respectively. As seen in FIG. 38B, mounting arm 1256A can have a plurality of holes 1257A to provide adjustment for the distance of press wheels 255A along the direction of travel.

(158) As viewed in FIGS. 39 to 43, also connected to press wheel frame 1250A is arm 1290A. Arm 1290A connects with brackets 1214A-1 and 1214A-2. Brackets 1214A-1 and 1214A-2 are connected to actuator base 1213A (as seen in FIGS. 32 and 33). As force is applied from actuator 259 through actuator base 1213A, force is applied to both the closing wheels 254 and press wheels 255A. Arm 1290A is also connected to arms 1295A-1 and 1295A-2. Arms 1295A-1 and 1295A-2 are disposed about pivots 1206A-1 and 1206A-2, respectively. Crossbar 1260A is connected to arms 1295A-1 and 1295A-2 and is disposed transverse to the direction of travel. As seen in FIG. 36, crossbar 1260A has a first end 1261A-1 disposed in opening 1225A-1 of swing arm 1220A-1 and a second end 1261A-2 disposed in opening 1225A-2 of swing arm 120A-2.

(159) In another embodiment, an angle sensor, which is described in International Publication No. WO2017/197274 as angle sensor 3006, or in

International Publication No. WO2014/066650 as angle sensor **280**, can be included. As illustrated in FIGS. **39** and **41**, the angle sensor is a Hall effect sensor **1400A** and a magnet **1401A**. While it can be installed on either the right or left side, Hall effect sensor **1400A** is illustrated as being disposed on second bracket **1204A-2** adjacent to arm **1295A-2**. A magnet **1401A** is disposed on arm **1295A-2**. In this embodiment, the angular rotation of arm **1295A-2**, which is connected to the entire assembly of the closing wheels **254** and press wheels **255A**, is measured as the average of both closing wheels **254** and both press wheels **255A**.

(160) Optionally, a bracket **1299** can be included to route a line (not shown) through. The line could be for applying material into the furrow, such as fertilizer, herbicide, or insecticide. Bracket **1299** can be connected to crossbar **1260A** as seen in FIG. **40**.

(161) As seen in FIG. **41**, there is an arm **1296A** pivotably disposed between brackets **1214A-1** and **1214A-2** about pivot **1293A** and between first bracket **1204A-1** and second bracket **1204A-2** about pivot **1295A**.

(162) Optionally, a trailing arm (not shown), such as the bracket **132** and flap **130** from International Publication No. WO2014/066650, can be included. The optional trailing arm can be connected to mounting arm **1256A** at connection **1288A**.

(163) Optionally, toe angle shim **1290A** (**1290A-1** and **1290A-2**) can be included to change the toe angle of closing wheels **254A** (**254A-1**, **254A-2**) by being disposed over axle **1291A** (**1291A-1**, **1291A-2**), respectively.

(164) The trench closing assembly **250** or **250A** can be a single stage by not including press wheels **255**, **255A** and the associated press wheel frame **1250**, **1250A**.

(165) In any of the embodiments, the distribution of force between the closing wheels **254** and press wheels **255** can be adjusted. In some embodiments, 80% of the force applied by actuator **259** is applied to the closing wheels **254** and 20% to the press wheels **255**. In another embodiment, 100% of the force can be applied to the closing wheels **254**.

(166) FIGS. **47** and **48** illustrate a trench closing assembly **250B** according to another embodiment. In this embodiment, trench closing assembly **250B** has a main frame **251B** that is pivotably connected to row unit **200**. Disposed on row unit frame **204** is bracket **249** that extends over main frame **251B**. Disposed between bracket **249** and main frame **251B** is actuator **259** to apply a variable force to trench closing assembly **250B**. Closing wheels **254** (**254-1** and **254-2**) are disposed on main frame **251B**. Pivotably connected to main frame **251B** is secondary frame **252B**. Secondary wheels **255** (**255-1**, **255-2**), such as press wheels, are disposed on secondary frame **252B**. Extending upward from main frame **251B** is bracket **8251**, and extending upward from secondary frame **252B** is bracket **8252**. Connecting brackets **8251** and **8252** is depth adjuster **253B**. The relative angle between main frame **251B** and secondary frame **252B** is adjusted with depth adjuster **253B**. This allows secondary wheels **255** to act as gauge wheels for closing wheels **254**.

(167) FIGS. **50-63** illustrate a trench closing assembly **250C** according to another embodiment. In this embodiment, trench closing assembly **250C** has a main frame **251C** that is pivotably connected to row unit **200**. Trench closing assembly **250C** has a frame **251C**, an actuator **259**, a pair of closing wheels **254-1**, **254-2**, and optionally, a press wheel **255C**. As illustrated, the press wheel **255C** may comprise a pair of press wheels **255C-1**, **255C-2**, but a single press wheel (not shown) may be utilized. Actuator **259** can apply one force to frame **251C**, and this force can be divided between the closing wheels **254** and the press wheel **255C**.

(168) Referring to FIG. **50**, frame **251C** has a main frame **1200C** having a connection bracket **1201C** that is adapted to be connected to row unit **200**. As in a previous embodiment, one or more bolts **1209** (FIG. **33**) can extend through apertures **1207** in connection bracket **1201C** for mounting the frame **251C** to the row unit **200**. Extending up from connection bracket **1201C** is actuator bracket **1210C**. In one embodiment, actuator bracket **1210C** has a first bracket arm **1211C-1**, a second bracket arm **1211C-2**, and a cross connector **1212C** connecting first bracket arm **1211C-1** and second bracket arm **1211C-2**. Actuator bracket **1210C** can be made from separate parts or as a single part. As best viewed in FIGS. **55**, **58**, and **65**, extending downwardly from connection bracket **1201C** are first bracket **1204C-1** and second bracket **1204C-2** with cross-braces **1208C-1**, **1208C-2**, and **1208C-3** extending first bracket **1204C-1** and second bracket **1204C-2**. Extending outward, transverse to the direction of travel, are optional stops **1229C**. Stop **1229C-1** is disposed on first bracket **1204C-1**, and stop **1229C-2** is disposed on second bracket **1204C-2**. Stop **1229C-1** cooperates with stops **1228C** (**1228C-1a** and **1228C-1b**) on first swing arm **1220C-1**. Stop **1229C-2** cooperates with stops **1228C** (**1228C-2a** and **1228C-2b**) on second swing arm **1220C-2**. The angle of rotation of swing arms **1220C** can be limited.

(169) As viewed in FIG. **61**, extending through first bracket **1204C-1** is pivot **1206C-1** transverse to the direction of travel. Extending through second bracket **1204C-2** is pivot **1206C-2** transverse to the direction of travel. Pivots **1206C-1** and **1206C-2** allow for pivoting of swing arms **1220C-1** and **1220C-2**, respectively.

(170) Swing arms **1220C-1** and **1220C-2** are pivotably disposed about pivots **1206C-1** and **1206C-2**, respectively. Swing arms **1220C-1** and **1220C-2** are adjustable transverse to the direction of travel along pivot **1206C-1** and pivot **1206C-2**, respectively. This allows for changing the width of spacing of closing wheels **254**.

(171) As seen in FIG. **61**, an actuator base **1213C** is connected to first base arm **1214C-1** and second base arm **1214C-2**. First base arm **1214C-1** and second base arm **1214C-2** are pivotably connected to pivot **1223**. Also first transfer arm **1287-1** and second transfer arm **1287-2** are disposed about pivot **1223**. Disposed forward of pivot **1223** in a direction of travel is pivot **1224**. Pivot **1224** is disposed through swing arm **1220C-1** and swing arm **1220C-2**, and first transfer arm **1287-1** and second transfer arm **1287-2** are pivotably disposed about pivot **1224**. Disposed rearward of pivot **1223** in a direction of travel is pivot **1226**. Pivot **1226** is disposed through swing arm **1220C-1** and swing arm **1220C-2**, and first transfer arm **1287-1** and second transfer arm **1287-2** are pivotably disposed about pivot **1226**. Also disposed about pivot **1226** are first bracket **1284-1** and second bracket **1284-2**. Disposed between first bracket **1284-1** and second bracket **1284-2** is cross brace **1264**. As viewed in FIG. **63**, cross brace **1264** has a hole **1265** for bolt **1273** to be disposed through. As viewed in FIG. **53**, mounting arm **1256C** is connected between first bracket **1284-1** and second bracket **1284-2**. Mounting arm **1256C** has the same design as a previous embodiment. An arm **1296C** is disposed about pivot **1293C** between first base arm **1214C-1** and second base arm **1214C-2**. Arm **1296C** is also disposed about pivot **1222** between first bracket **1204C-1** and second bracket **1204C-2**.

(172) As seen in FIGS. **56-63**, an adjuster **1272** is included for changing the relative position of closing wheels **254** (**254-1** and **254-2**) to press wheels **255C** (**255C-1** and **255C-2**). By changing the relative position, the percentage of the applied force from actuator **259** is changed between the closing wheels **254** (**254-1** and **254-2**) the press wheels **255C** (**255C-1** and **255C-2**). Adjuster **1272** has a bracket **1276** having a first arm **1279-1** a second arm **1279-2**, a cross-connector **1277**, and a hole **1278** through cross-connector **1277**. Bracket **1276** can be made as a unitary part or from separate parts. First arm **1279-1** and second arm **1279-2** are pivotably disposed about pivot **1227**. Pivot **1227** is disposed through swing arm **1220C-1** and swing arm **1220C-2**. Disposed through hole **1278** is bolt **1273**. Bolt **1273** is also disposed through hole **1265** in cross brace **1264**. Knob **1275** is disposed about bolt **1273** to adjust the position of bolt **1273**. Optionally, a retaining nut **1274** can be disposed at the end of bolt **1273** to retain knob **1275** on bolt **1273**. Optionally, force sensor **1298** can be disposed between knob **1275** and cross brace **1264** about bolt **1273**. The combined down force applied to closing wheels **254** (**254-1** and **254-2**) and press wheels **255C** (**255C-1** and **255C-2**) can be measured by force sensor **1298**. An example of force sensor **1298** is a load sensor, such as the Case IH Load Sensor, Part No. 725875, from Precision Planting, LLC.

(173) As bolt **1273** is adjusted upwards by knob **1275**, pivot **1273** is pulled upwards by first arm **1279-1** and second arm **1279-2**, which raises swing arms **1220C** (**1220C-1** and **1220C-2**) under adjuster **1272**. As bolt **1273** is adjusted downwards by knob **1275**, pivot **1273** is lowered by first arm **1279-1** and second arm **1279-2**, which lowers swing arms **1220C** (**1220C-1** and **1220C-2**) under adjuster **1272**.

(174) Wheel

(175) FIGS. **21-24** illustrate a wheel **1300** according to one embodiment. Wheel **1300** can be used as press wheel **255**, **255A** or **255C**. Wheel **1300** comprises a spoke disk **1310** having a hub **1301** and a plurality of spokes **1302** (**1302-1** to **1302-12**) radially disposed about hub **1301**. Disposed at the end of a spoke **1302** is a cleat **1303** (**1303-1** to **1303-12**). Each spoke **1302** can be a two part spoke. FIG. **23** illustrates the spoke disk **1310** having a plurality of spoke arms **1311** (**1311-1** to **1311-12**). Each spoke arm **1311-1** to **1311-12** has a leading edge **1314** and a trailing edge **1315**. The radial

end of each spoke arm **1311-1** to **1311-12** to **1311-1** to **1312-1** to **1312-12** is a respective flange **1313-1** to **1312-12**. FIG. 24 is a perspective view of the spoke disk **1310** showing flanges **1313-12** and **1313-3** disposed in the respective notches **1312-12** and **1312-3** of spoke arms **1311-12** and **1311-3**. As best viewed in FIGS. 21 and 22, a respective cleat **1303-1** to **1303-12** is attached to each of the respective flanges **1313-1** to **1313-12**. As best viewed in FIG. 22, the cleat **1303** can have an L shape. Referring to FIG. 21, a forward edge **1304** of each of the cleats **1303** is shown aligned with the leading edge **1314** of the respective spoke arm **1311**, with a rearward end **1305** of each cleat **1303** extending rearwardly of the trailing end **1315** of the spoke arm **1311** substantially spanning the space between the adjacent radial spokes **1302**.

(176) In one aspect, wheel **1300** can move soil from the sides of a seed trench to knit the soil together to increase the amount of closing of the seed trench.

(177) FIGS. 44-46 illustrate another embodiment of a wheel **1300A**. Wheel **1300A** can be used as press wheel **255**, **255A** or **255C**. Wheel **1300A** comprises a spoke disk **1310A** and a hub **1301A**. Spoke disk **1310A** can be molded as a unitary part. Spoke disk **1310A** has plurality of spokes **1302A** (**1302A-1** to **1302A-10**). Connecting spokes **1302A** is a tread **1320A**. Tread **1320A** has a rib **1315A** (**1315A-1** to **1315A-10**) disposed at the radial end of spoke **1302A**. Between each rib **1315A**, there is a tread portion **1316A** (**1316A-1** to **1316A-10**). Tread portion **1316A** can extend through the entire width **W** of a rib **1315A**, or tread portion **1316A** can extend only a portion of the width **W** to leave a gap **1317A** (**1317A-1** to **1317A-10**).

(178) FIGS. 117-124 illustrate wheel **1300'**. The wheel **1300'** may be used in place of a press wheel **255** (**255**, **255A**, **255C**, **255D**) or the wheel **1300** (**1300**, **1300A**, **1300B**), which are described above and in U.S. Provisional Patent Application No. 62/731,813 and International Application No. PCT/US2019/020452, both of which are incorporated herein in their entirety by reference. In FIG. 17, the wheel **1300'** (**1300'-1**, **1300'-2**) is shown disposed on a trench closing assembly **250D**, describe above.

(179) Turning to FIGS. 118-124, wheel **1300'** has a central axis and a circumferential tread **1320'** coaxial with the central axis. The circumferential tread **1320'** extends axially between an inner edge and an outer edge defining an axial width **W'** (FIG. 123) of the wheel **1300'**. The tread **1320'** may include a plurality of ribs **1315'** (**1315'-1** to **1315'-10**). While the wheel **1300'** is shown with **10** ribs **1315'** it should be appreciated that any number of ribs may be used. In one embodiment, the ribs **1315'** extend axially in a direction substantially parallel with the central axis. Between each rib **1315'** there is a tread web **1316'** (**1316'-1** to **1316'-10**). The tread web **1316'** may extend through the entire axial width **W'** between each of the ribs **1315'**. Alternatively, the tread web **1316'** may extend only a portion of the axial width **W'** between the ribs **1315'**, such as illustrated in the embodiments of wheels **1300A** and **1300B**.

(180) The wheel **1300'** includes a wall member **1330'** joined with the circumferential tread **1320'**. The wall member **1330'** extends radially inward from the circumferential tread **1320'** to a central opening **1333'** coaxial with the central axis. The wall member **1330'** is continuous (i.e., solid) between the central opening **1333'** and the circumferential tread **1320'** for the reasons discussed later. The central opening **1333'** receives a hub **1301'** by which the wheel **1300'** may be rotatably mounted to the frame of the trench closing assembly **250**. The wall member **1330'** has a first side **1331'** (outer side) and a second side **1332'** (inner side). The wall member **1330'** extends axially outward from the inner edge of the circumferential tread **1320'** toward the outer edge of the circumferential tread **1320'** by an axial distance **D'** (FIG. 123), but the axial distance **D'** does not extend outward beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width **W'**. In one embodiment, the axial distance **D'** is equal to or less than half of the axial width **W'**. In another embodiment, the axial distance **D'** is less than one half of the axial width **W'**. In another embodiment, the axial distance **D'** is equal to or greater than 10% of the axial width **W'**. As seen in FIG. 123, when viewed in cross-section, the circumferential tread together with the wall **1330'** has a cross-sectional shape in the form of an uppercase Greek letter sigma (**X**). In one embodiment, by having the axial distance **D'** not extend beyond the vertical plane perpendicular to the central axis at a midpoint of the axial width, the portion of the tread **1320'** that extends axially outward beyond the first side **1331'** of the wall member **1330'** is able to flex radially inward towards the central axis of the wheel **1300'**. When the hub **1301'** is installed, the center of the wheel **1300'** is solid so that nothing can pass through the wall member **1330'** from the second side **1332'** to the first side **1331'** or vice versa. Thus, when in muddy conditions, the solid wall of the wheel **1300'** prohibits a flow of mud from one side of the wheel to other. It should be appreciated that if the wall member **1330'** is spoked or has openings through the wall **1330'** from one side of the wheel to the other, mud may become entrapped, which may require cleaning of the wheel to remove the mud.

(181) FIGS. 125-130 illustrate another embodiment of a wheel **1300''** which may be used in place of a press wheel **255** (**255**, **255A**, **255C**, **255D**) or any of the wheels **1300**, **1300A**, **1300B**, **1300'** referenced above. Similar to wheel **1300'**, wheel **1300''** has a central axis and a circumferential tread **1320''** coaxial with the central axis. The circumferential tread **1320''** extends axially between an inner edge and an outer edge defining an axial width **W''** (FIG. 130) of the wheel **1300''**. The tread **1320''** may include a plurality of ribs **1315''** (**1315''-1** to **1315''-10**). While the wheel **1300''** is shown with **10** ribs **1315''** it should be appreciated that any number of ribs may be used. In one embodiment, the ribs **1315''** extend axially in a direction substantially parallel with the central axis. Between each rib **1315''** there is a tread web **1316''** (**1316''-1** to **1316''-10**). The tread web **1316''** may extend through the entire axial width **W''** between each of the ribs **1315''**. Alternatively, the web tread **1316''** may extend only a portion of the axial width **W''** between the ribs **1315''**, such as illustrated in the embodiments of wheels **1300A** and **1300B**.

(182) The wheel **1300''** includes a wall member **1330''** joined with the circumferential tread **1320''**. The wall member **1330''** extends radially inward from the circumferential tread **1320''** to a central opening **1333''** coaxial with the central axis. The wall member **1330''** is continuous (i.e., solid) between the central opening **1333''** and the circumferential tread **1320''** for the reasons discussed later. Similar to the previously described wheel **1300'**, the central opening **1333''** of the wheel **1300''** is adapted to receive a hub (not shown) by which the wheel **1300''** may be rotatably mounted to the frame of the trench closing assembly **250**. The wall member **1330''** has a first side **1331''** (outer side) and a second side **1332''** (inner side). The wall member **1330''** extends axially outward from the inner edge of the circumferential tread **1320''** toward the outer edge of the circumferential tread **1320''** by an axial distance **D''** (FIG. 130), but the axial distance **D** does not extend axially outward beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width **W''**. In one embodiment, the axial distance **D''** is equal to or less than half of the axial width **W''**. In another embodiment, the axial distance **D''** is less than one half of the axial width **W''**. In another embodiment, the axial distance **D''** is equal to or greater than 10% of the axial width **W''**. As seen in FIG. 130, when viewed in cross-section, the circumferential tread together with the wall **1330''** has a cross-sectional shape in the form of an uppercase Greek letter sigma (**X**). In one embodiment, by having the axial distance **D''** not extend beyond the vertical plane perpendicular to the central axis at a midpoint of the axial width, the portion of the tread **1320''** that extends axially outward beyond the first side **1331''** of the wall member **1330''** is able to flex radially inward towards the central axis of the wheel **1300''**. When the hub is installed, the center of the wheel **1330''** is solid so that nothing can pass through the wall **1330''** from the second side **1332''** to the first side **1331''** or vice versa. Thus, when in muddy conditions, the solid wall of the wheel **1300''** prohibits a flow of mud from one side of the wheel to other. It should be appreciated that if the wall member **1330''** is spoked or has openings through the wall **1330''** from one side of the wheel to the other, mud may become entrapped, which may require cleaning of the wheel to remove the mud.

(183) Pressure Control

(184) To control the flow of fluid (e.g., air) to actuator **259**, a control valve **258** can be included. As illustrated in FIG. 25 for an eight row system, there can be a control valve **258** (**258-1** to **258-8**) associated with each actuator **259** (**259-1** to **259-8**). A fluid supply line **257** can supply fluid (e.g., air) from a fluid supply system. While shown schematically, control valve **258** can be disposed on the row unit **200**, trench closing assembly **250**, or on toolbar **202**. Different locations are shown in FIG. 25 for illustrating different locations, but they can all be the same. Section control can also be used. FIG. 26 illustrates section control in which a control valve **258** (**258-9** to **258-12**) supplies fluid to two or more actuators **259** (**259-1** to **259-8**). While shown with one control valve **258** to two actuators **259**, any amount of sectioning can be used up to having one control valve **258** supplying all actuators **259** (not shown). Each control valve **258** can be in signal communication with monitor **300** for controlling each valve **258**. While illustrated for air, which can be vented to atmosphere, the fluid can be hydraulic, which can further include a return line (not shown). An example of control

valve **258** is the ITV series (such as ITV **1051**) electro-pneumatic valve from SMC Pneumatics. This electro-pneumatic valve has a solenoid supply valve for inlet air and a solenoid valve for exhaust to atmosphere. When air is needed to the actuator, the inlet air valve is open and the valve to atmosphere is closed. When air pressure at the actuator needs to be reduced, the inlet air valve is closed, and the valve to atmosphere is opened. (185) While a single control valve **258**, such as the ITV **1051** valve, can be used, an equivalent valve system **258A** can be used that is made from component parts. Valve system **258A** is illustrated in FIG. **49**. Valve system **258A** is supplied with fluid (e.g., air) from line **9901** to inlet valve **9905**. Inlet valve **9905** discharges to line **9903**, which is connected to actuator **259**, exhaust valve **9910**, and pressure sensor **9906**. Pressure sensor **9906** is in signal communication with a circuit **9907**, which is in signal communication with monitor **300**. Exhaust valve **9910** discharges to atmosphere, such as through optional line **9902**.

(186) As is described in International Publication No. WO2017/197274, control of the amount of force to actuator **259** can be based on input from one or more of the trench closing sensor, the angle sensor, a force sensor **261**, which is disposed on trench closing assembly **250**, or position sensor **900**. The control can be closed loop or open loop. Force sensor **261** can be disposed on trench closing assembly **250** at any location that measures the force on any part of trench closing assembly **250**. In one embodiment, force sensor **261** is disposed to measure a force applied to press wheels **255**. An example of a location is at location **9261-A** or location **9261-B**, which is illustrated in FIG. **27**. For location **9261-A**, a load sensing pin can be used. For location **9261-B**, a Wheatstone bridge can be used.

(187) For packing/firming wheels in general, a force sensor **261** can be a load pin installed on the axle, on the arm connecting the packing/firming wheel to the trench closing assembly **250** or row unit **200**, at the connection of the packing/firming wheel arm to trench closing assembly **250** or row unit **200**, or where a spring/actuator connects to the packing/firming wheel frame. FIGS. **66-69** illustrate various locations for force sensors **261** on different press wheel systems **9255A**, **9255B**, **9255C**, and **9255D**. Each press wheel system **9255A**, **9255B**, **9255C**, and **9255D** has a press wheel **255D** and mounting arms **9256-1** and **9256-2**. Press wheel system **9255A**, **9255B**, **9255C** and **9255D** have a connection bracket **9259** for connection to row unit **200** or to closing system **250**. A load sensor **9258** can be disposed between connection bracket **9259** and row unit **200** or closing system **250**. A load sensing pin **9257** can be disposed on axle **9253** or as shown in FIG. **67** on axles **9253-1** or **9253-2**. A force sensor **9254**, such as a Wheatstone bridge, can be disposed on arm **9256** (**9256-1** or **9256-2**) or as shown in FIG. **67** on arm **9256** or shank **9260**. In FIGS. **67** and **68**, a force sensor **9262** can be disposed where spring **9250** is attached to frame **9251** or **9251A**. In FIG. **66**, a force sensor **9262** can be disposed at connection **9249**.

(188) Force sensors **261**, **9258**, **9257**, **9254**, **9262** are in signal communication with monitor **300**.

(189) In another embodiment, instead of a force sensor **261**, a position sensor **900** can be used. Referring to FIG. **28**, modified mounting arm **1256A** has a first section **1256A-1** and a second section **1256A-2** connected with hinge **901** and with biasing member **902** (such as a spring) disposed between first section **1256A-1** and second section **1256A-2**. Position sensor **900** includes a transmitter **904** and a receiver **903**. It is understood that the positions of transmitter **904** and receiver **903** can be switched. An example of the transmitter **904** and receiver **903** are a magnet and Hall effect sensor. As downward forces are applied to modified mounting arm **1256A**, biasing member **902** is compressed, and the distance between transmitter **904** and receiver **903** is reduced. Position sensor **900** is in signal communication with monitor **300**.

(190) Soil Leveler

(191) FIGS. **70-74** illustrate a soil leveler **8000** according to one embodiment. As illustrated, soil leveler **8000** is attached to mounting arm **1256C** of trench closing assembly **250C**. Soil leveler has a first bracket **8020-1** and a second bracket **8020-2** for attaching to mounting arm **1256C**. Pivotably connected to first bracket **8020-1** and second bracket **8020-2** through pivot **8030** are first arm **8010-1** and second arm **8010-2**. Optionally, to stabilize first arm **8010-1** and second arm **8010-2** is cross brace **8015** connecting first arm **8010-1** and second arm **8010-2**. To adjust a relative angle between the first arm **8010-1** and second arm **8010-2** and first bracket **8020-1** and second bracket **8020-2**, each arm **8010** (**8010-1** and **8010-2**) have a slot **8011** (**8011-1** shown, and **8011-2** not shown). A fastener **8019** is disposed through slots **8011** and through first bracket **8020-1** and second bracket **8020-2**. Each arm **8010** (**8010-1** and **8010-2**) have a space **8015** (**8015-1** and **8015-2**) having notches **8016** (**8016-1** and **8016-2**), respectively. Plate **8045** is engagable with notches **8016** (**8016-1** and **8016-2**). Attached to plate **8045** and disposed about fastener **8019** is biasing element **8040** (such as a spring). Moving plate **8045** to different notches **8016** (**8016-1** and **8016-2**) adjusts the amount of biasing force applied to arms **8010** (**8010-1** and **8010-2**). Attached to arms **8010** (**8010-1** and **8010-2**) is flap **8060**. Flap **8060** can be a unitary part, or as illustrated, flap **8060** has plate **8061** attached to arms **8010** (**8010-1** and **8010-2**) and flap portion **8065** attached to plate **8061**. Optionally, flap portion **8065** can have a serrated edge **8066** for engaging the soil. Serrated edge **8066** can be sloped upwards from edges **8062** (**8062-1**, **8062-2**) to a center **8063** of flap portion **8065**.

(192) FIGS. **75-79** illustrate another soil leveler **8000A**. As illustrated, soil leveler **8000A** is attached to mounting arm **1256C** of trench closing assembly **250C**. Soil leveler **8000A** has a first bracket **8120-1** and second bracket **8120-2** for attaching to mounting arm **1256C**. Connected to first bracket **8120-1** and second bracket **8120-2** are first arm **8110-1** and second arm **8110-2**. As shown, first arm **8110-1** and second arm **8110-2** are disposed between first bracket **8120-1** and second bracket **8120-2**, but first arm **8110-1** and second arm **8110-2** are disposed outside of first bracket **8120-1** and second bracket **8120-2**. Arms **8110** (**8110-1** and **8110-2**) have a first section **8112** (**8112-1** and **8112-2**) for connecting to first bracket **8120-1** and second bracket **8120-2**. Extending laterally outward (transverse to the direction of travel) is a lateral segment **8111** (**8111-1** and **8111-2**). Disposed downwardly from lateral segment **8111** (**8111-1** and **8111-2**) is leg **8113** (**8113-1** and **8113-2**). Disposed in leg **8113** (**8113-1** and **8113-2**) is a hole **8114** (**8114-1** and **8114-2**). While not shown, a drag line extends from hole **8114-1** to hole **8114-2** and has a length to drag across the ground. Examples of the drag line include, but are not limited to, a chain, a wire, a cable, or a rope. Each bracket **8120** (**8120-1** and **8120-2**) have an adjustment slot **8121** (**8121-1** and **8121-2**) through which fastener **8122** is disposed, and fastener **8122** is disposed through first section **8112** (**8112-1** and **8112-2**) of arms **8110** (**8110-1** and **8110-2**). Fastener **8123** is also disposed through each bracket **8120** (**8120-1** and **8120-2**) and each first section **8112** (**8112-1** and **8112-2**) of arms **8110** (**8110-1** and **8110-2**). The relative angle between the brackets **8120** (**8120-1** and **8120-2**) and the arm **8110** (**8110-1** and **8110-2**) can be adjusted by pivoting about fastener **8123**. To adjust a distance between holes **8114** (**8114-1** and **8114-2**) and brackets **8120** (**8120-1** and **8120-2**), first sections **8112** (**8112-1** and **8112-2**) can optionally have a plurality of holes **8115** for adjusting a position of arms **8110** (**8110-1** and **8110-2**).

(193) FIGS. **80-104** illustrate a trench closing assembly **250D** according to another embodiment. In this embodiment, trench closing assembly **250D** has a main frame **251D** that is connected to row unit **200**. Trench closing assembly **250D** has a frame **1520**, an actuator **259**, a pair of closing wheels **254D-1**, **254D-2**, and optionally, a press wheel **255D**.

(194) FIGS. **115** and **116** are enlarged perspective and side elevation views, respectively, of the closing wheels **254D-1**, **254D-2** of the trench closing assembly **250D**. The closing wheels **254D-1**, **254D-2** are mirror images of one other and therefore only one closing wheel is shown in FIGS. **115** and **116** generally designated by referenced by reference number **254D**. In this embodiment, the closing wheels **254D** comprise a generally dish-shaped body **510** in the form of a disc blade with a circumferential edge **512** and having a concave surface **514** on one side and a convex surface **516** on the opposing side. As best viewed in FIG. **80**, the closing wheels **254D-1**, **254D-2** are mounted so as to be disposed on each side of the open furrow with their convex sides **516** oriented inward toward the open furrow. The closing wheels **254D-1**, **254D-2** are also oriented such that they angle outwardly upward (i.e., their respective circumferential edges **512** are closer to one another or they converge toward one another in the direction of the soil surface). Additionally, the closing wheels **254D-1**, **254D-2** are mounted at an angle with respect to one another fore and aft (i.e., in the direction of travel) such that their respective circumferential edges **512** are closer to one another toward the rear than toward the forward direction of travel. Thus, due to their orientation and the convex shape toward the open furrow, as the row unit is drawn through the field causing the closing wheels to rotate through the soil, the closing wheels act to push the soil inward toward the open furrow, thus “closing” or filling the furrow with soil to cover the previously deposited seed.

(195) The circumferential edge **512** of the dish-shaped body **510** may be continuous or the circumferential edge **512** may include a series of radially

paced notches **518** that are cut or otherwise formed around the outer circumferential periphery resulting in a series of radially spaced teeth or spikes **520**. The notches **518** may be formed, so that each tooth or spike **520** curves or sweeps opposite the direction of rotation as indicated by arrow **511**. It should be appreciated that this rearward swept orientation will reduce the amount of soil thrown by the teeth **520** because the teeth rotate out of the soil more vertically than if the teeth were straight. To improve penetration into the soil, the convex side **516** may be ground down, beveled or otherwise formed to taper toward the circumferential edge **512** as best viewed in FIG. **116**, so the circumferential edge **512** is thinner or sharper for easier penetration into the soil.

(196) The dish shaped body **510** may have a generally flat or planar central region **522** with bolt holes **524** and a central aperture **526** for mounting to a hub or spindle as shown in FIG. **80**.

(197) The closing wheel **524D** may be fabricated by any suitable means. One exemplary manner of fabrication is to cut the body **510** out of flat plate steel having a generally uniform thickness to produce a wheel blank. The blank may then be placed in a forming die and pressed to the desired dish-shape. The dish-shaped blank may then undergo further processing, such as cutting the notches **518**, if desired, to form teeth **520** around the outer circumferential periphery. The outer circumferential periphery may then be ground or tapered to produce a thinner or sharper outer circumferential edge. Alternatively the notches or teeth may be cut into the blank before being pressed to the desired dish shape.

(198) As illustrated, the press wheel **255D** may comprise a pair of press wheels **255D-1**, **255D-2**, but a single press wheel (not shown) may be utilized. Actuator **259** can apply one force between frame **251D** and frame **1520**, and this force can be divided between the closing wheels **254** and the press wheel **255D**.

(199) Turning to FIGS. **91-96**, frame **251D** has a connection bracket **1540** and an attachment bracket **1550**. Connection bracket **1540** can have one or more bolts **1209D** extending through apertures **1207D** in connection bracket **1540** for mounting the frame **251D** to the row unit **200**. Connection bracket **1540** (as with connection brackets **1200** and **1201A**) can be varied to mate attachment to different styles of row units. Connection bracket **1540** has a first side **1542-1**, a second side **1542-2**, and a plate **1543** disposed between first side **1542-1** and second side **1542-2**. First side **1542-1** and second side **1542-2** each have a post **1541-1** and **1541-2**, respectively, protruding perpendicularly outward.

(200) Attachment bracket **1550** connects to connection bracket **1540**. Attachment bracket **1550** provides a common structure for mounting other parts while connection bracket **1540** has a varied structure to mate with different styles of row units. Attachment bracket has a first side **1552-1**, a second side **1552-2**, a crossbar **1557** disposed between first side **1552-1** and second side **1552-2**, and plate **1553** disposed between first side **1552-1** and second side **1552-2**, and plate **1553**. First side **1552-1** and second side **1552-2** each have a u-shaped opening **1555-1** and **1555-2**, respectively, for connection to posts **1541-1** and **1541-2**, respectively. First side **1552-1** and second side **1552-2** have openings **1558-1** and **1558-2**, respectively, for accepting pivots **1529-1** and **1529-2**, respectively. Attachment bracket **1550** can be secured to connection bracket with fastener **1549**. Optionally, attachment bracket **1550** can also have an opening **1501** disposed in first side **1552-1** or second side **1552-2** for accepting a pin. While connection bracket **1540** is illustrated with separate parts it may be fabricated as a unitary part.

(201) Optionally, as illustrated in FIGS. **82** and **84**, a guard **1590** can be disposed on frame **1520** ahead of closing wheels **254D-1**, **254D-2**. Guard **1590** can prevent rocks, rootballs, or other trash from approaching closing wheels **254D-1**, **254D-2**. The height of guard **1590** can be adjusted by changing the placement of arms **1591-1** and **1591-2** on frame **1520**.

(202) Turning to FIGS. **97-98**, mounting arm **1570** has a first side **1571-1**, a second side **1571-2**, a plate **1574** disposed between first side **1571-1** and second side **1571-2**, a hole **1573** through plate **1574** for passage of sleeve **1603**, axle hubs **1575-1** and **1575-2** for mounting press wheels **255D-1** and **255D-2**, respectively, and, optionally, a handle cradle **1579**.

(203) Turning to FIGS. **83-90**, frame **1520** is pivotally connected to frame **251D** through pivots **1529-1** and **1529-2**. Frame **1520** has a first side **1526-1** and a second side **1526-2**, a plate **1525** connecting first side **1526-1** and second side **1526-2**, a connection arm **1527** extending rearwardly along the direction of travel from plate **1525**, a cross-brace **1528-1** connecting connection arm **1527** to first arm **1526-1**, a cross-brace **1528-2** connecting connection arm **1527** to second arm **1526-2**. First side **1526-1** and second side **1526-2** have openings **1521-1** and **1521-2** for disposing about pivots **1529-1** and **1529-2**, respectively. Mounting arm **1570** can attach to frame **1520** at connections **1522-1a** and **1522-2a** or to **1522-1b** and **1522-2b**. The plurality of connections allows the distance between closing wheels **254D-1**, **254D-2**, and press wheel **255D** (**255D-1**, **255D-2**) to be changed. There can be one connection **1522** or a plurality of connections **1522**. As the mounting arm **1570** position is changed, there are also corresponding connections **1523-a** and **1523-b** (matching in number to connection **1522**) for connecting handle assembly **1600** to frame **1520**. Frame **1520** also has connections **1524** (**1524-1a**, **1524-1b**, **1524-2a**, and **1524-2b**) for mounting closing wheels **254D-1**, **254D-2**. While there can be one set of connections **1524**, the plurality of connections **1524** allow for forward and back placement of the closing wheels **254D-1**, **254D-2** on frame **1520**, or closing wheels **254D-1**, **254D-2** can be offset from each other with one closing wheels **254D-1**, **254D-2** being mounted to a forward location (the “a” position) or to a rear location (the “b” position). As illustrated, closing wheels **254D-1**, **254D-2** are offset from each other. Optionally, frame **1520** can have an opening **1502** in either the first side **1526-1** or second side **1526-2** (shown in **1526-2**) for accepting a pin. Frame **1520** can be raised to allow opening **1501** and opening **1502** to align for accepting a pin (not shown). This allows trench closing assembly **250D** to be raised for transport or when closing is not needed. While illustrated on one side, openings **1501** and **1502** can be disposed on both sides.

(204) Actuator **259** is disposed between plate **1525** and plate **1553** to apply a force to plate **1553** to cause frame **1520** to pivot and apply pressure to closing wheels **254D-1** and **254D-2**.

(205) Turning to FIGS. **99-104**, handle assembly **1600** is illustrated. Handle assembly **1600** has a sleeve **1603** having a first diameter **1620** and a second diameter **1621**. Second diameter **1621** is small enough to be disposed through load sensor **1610**, and first diameter **1620** is large enough so that it cannot pass through load sensor **1610**. Sleeve **1603** has a bracket **1622** (u-shaped bracket as illustrated or any other shape) for accepting handle **1609**. Disposed on sleeve **1603** below bracket **1622** are bevel washers **1601** and **1602**. Bevel washer **1601** and bevel washer **1602** are disposed with their concave surfaces facing each other. This allows bevel washers **1601** and **1602** to flex to absorb shocks experienced by trench closing assembly **250D** to prevent overloading of load sensor **1610**. Before bevel washers **1601** and **1602** reach maximum flex, first diameter **1620** will contact plate **1604** to limit the travel. Load sensor **1610** is a pancake load sensor. Load sensor **1610** has a hole **1612** for passage of sleeve **1603**. Disposed on the underside of load sensor **1610** are a plurality of feet **1611** to allow load sensor **1610** to flex and measure force. Load sensor **1610** can be disposed on mounting arm **1570** directly, or as shown, a plate **1604** can be disposed between load sensor **1610** and mounting arm **1570**. Optionally, a washer **1607** can be disposed about sleeve **1603** under mounting arm **1570**. Handle assembly **1600** is connected to frame **1520** at connection **1523-a** or **1523-b** with a bracket **1606** (u-shaped bracket) and a bolt **1605** connecting bracket **1606** with sleeve **1603**. In the horizontal position, handle **1609** locks handle assembly **1600** in place against mounting arm **1570**. In the vertical position, handle **1609** releases handle assembly from engagement with mounting arm **1570**. Bolt **1605** can be adjusted to set a vertical placement of mounting arm **1570** relative to frame **1520**.

(206) Load sensor **1610** can be connected to a network directly through a plug (not shown) having a CAN processor to allow direct communication over a CAN network. The CAN processor can communicate pressure readings and provide control signals over the CAN network. Alternatively, load sensor **1610** can be connected to a control module (either an on-row module, or a module controlling a plurality of row) to communicate pressure readings that are then processed by the control module.

(207) FIGS. **105-107** illustrate another embodiment of a wheel **1300B**. Wheel **1300B** can be used as press wheel **255**, **255A**, **255C** or **255D**. Wheel **1300B** comprises a spoke disk **1310B** and a hub **1301B**. Spoke disk **1310B** can be molded as a unitary part. Spoke disk **1310B** has plurality of spokes **1302B** (**1302B-1** to **1302B-10**). Connecting spokes **1302B** is a tread **1320B**. Tread **1320B** has a rib **1315B** (**1315B-1** to **1315B-10**) disposed at the radial end of spoke **1302B**. Between each rib **1315B**, there is a tread portion **1316B** (**1316B-1** to **1316B-10**). Tread portion **1316B** can extend through the entire width A of a rib **1315B**, or tread portion **1316B** can extend only a portion of the width A to leave a gap **1317B** (**1317B-1** to **1317B-10**). Wheel **1300B** can be similar to wheel **1300A**. Tread portion **1316B** can be flexible to allow tread portion **1316B** to deflect inward towards hub

1301B. To limit the amount of flex of tread portion **1316B**, stops **1318B** can be disposed radially outward from spoke disk **1310B** between spokes **1302B**. The height of stops **1318B** (as a percentage of the distance between spoke disk **1310B** and tread portion **1316B**) and the width of stops **1318B** (as a percentage of the distance between spokes **1302B**) can be varied to regulate the amount of flex permitted for tread portion **1316B** so that mud that is built up on tread portion **1316B** will come off of tread portion **1316B**.

(208) FIGS. **108** to **114** illustrate a fluid control assembly **6800** according to one embodiment. Fluid control assembly **6800** controls the flow of fluid to and from actuator **259**. Fluid control assembly has a housing **6801**. There is a fluid inlet port **6802** from a fluid source (not shown). In one embodiment, the fluid can be air, but other fluids can be used. There is an outlet port **6803** on housing **6801** to return fluid to the fluid source, or in the case of air, outlet port **6803** can vent to atmosphere. Fluid control assembly **6800** has a conduit **6804** on housing **6801** to provide fluid communication to actuator **259**. Fluid control assembly **6800** has a port **6809** disposed through housing **6801** to allow signal communication with communication port **6890**.

(209) Turning to FIGS. **109-111** with the housing **6801** removed, board **6891** can be seen having communication port **6890** disposed on board **6891**. Board **6891** contains circuitry (not shown) to control fluid control assembly **6800**. A first valve **6810** and a second valve **6820** are disposed on board **6891**. First valve **6810** and second valve **6820** can be identical. Examples of these valves are 2-way pneumatic valves from Asco Valve, Inc., 160 Park Avenue, Florham Park, NJ, 07932. These valves can operate normally closed.

(210) First valve **6810** has an inlet **6811**, which is in fluid communication with inlet port **6802**, and an outlet **6812**, which is in fluid communication with conduit **6804**. Second valve **6820** has an inlet **6821**, which is in fluid communication with conduit **6804**, and an outlet **6822**, which is in fluid communication with outlet port **6803**. Seals **6889**, such as o-rings, can seal inlet **6811**, outlet **6812**, inlet **6821**, and outlet **6822**.

(211) Also disposed in fluid control assembly **6800** is a pressure sensor **6830**, which is in data communication with board **6891**. While pressure sensor **6803** can be disposed anywhere in fluid control assembly **6800**, it is illustrated as being disposed on board **6891**. Pressure sensor **6830** is in fluid communication with conduit **6804** through conduit **6808**. An example of pressure sensor **6803** is a Honeywell board mounted pressure sensor.

(212) Turning to FIGS. **112-114**, fluid communication of conduit **6804** is illustrated. Conduit **6804** is in fluid communication outlet **6802** through port **6816**. Conduit **6804** is in fluid communication with pressure sensor **6830** through conduit **6808**. Conduit **6804** is in fluid communication with inlet **6821** through port **6825**. Outlet **6822** is in fluid communication with outlet port **6803** through port **6826**. Inlet port **6802** is in fluid communication with inlet **6811** through port **6815**. Seals **6889** can be disposed in ports **6815**, **6816**, **6825**, and **6826**.

(213) Communication port **6890** can be any port used in various types of signal/data communication. Examples include, but are not limited to, Computer Area Network (CAN) port, USB, Ethernet, or RS-232. All processing of signals and control can be done on board **6891**, or signals can be sent to monitor **300** or a remote controller (not shown) for processing and control to return a signal to fluid control assembly **6800** to control valves **6810** and **6820**. Closed loop control can be used to control the pressure in actuator **259** to a selected value that is set by an operator. The selected value can be a selected amount of pressure for actuator **259**, or the selected value can be a selected position for trench closing assembly **250**, **250A**, **250B**, **250C** or **250D**, such as from position sensor **900**. As the hardness of soil changes, more or less pressure is needed to maintain the same amount of closing. In harder soils, an increase in pressure may be needed to obtain the same amount of closing, and in softer soils, a decrease in pressure may be used.

(214) In another embodiment, pressure to actuator **259** can be controlled based on an amount of closing as measured by a trench closing sensor. Trench closing sensors are described in International Patent Publication No. WO2017/197274, or in U.S. Application Nos. 62/586,397; 62/640,279; and 62/672,993, all of which are incorporated herein by reference in their entireties.

(215) In operation, when valves **6810** and **6820** are closed, pressure in actuator **259** can be measured by pressure sensor **6830**. If additional pressure is needed in actuator **259**, valve **6810** can be opened (with valve **6820** closed) to place inlet port **6802** in fluid communication to conduit **6804** to actuator **259**. If too much pressure is in actuator **259**, valve **6820** can be opened (with valve **6810** closed) to place outlet port **6803** in fluid communication to conduit **6804** to actuator **259**.

(216) Fluid control assembly **6800** can be disposed anywhere on trench closing assembly **250**, **250A**, **250B**, **250C**, **250D** or on row unit **200**.

(217) In another embodiment, a feed forward control method is provided to adjust the downforce control system **214** based on a force applied to the actuator **259**. If a change in force applied by the actuator **259** is applied, the same amount of change or a portion thereof can also be made to the downforce control system **214**. This can be done to balance the force on the row unit **200**. For example, if an increase of 50 units of force is to be applied by the actuator **259**, then a signal can be sent to also increase the force applied by the downforce control system **214** by an additional 50 units of force. In other embodiments, the change of force to the downforce control system **214** can be the opposite to the change in force applied by the actuator **259**. For example, if an increase of 50 units of force is to be applied by the actuator **259**, then a signal can be sent to decrease the force applied by the downforce control system **214** by 50 units of force. In either embodiment, the amount of change to the downforce control system **214** can be less than, equal to, or greater than the change in force applied by the actuator **259**.

(218) In another embodiment, there can be just one sensor **10000** per row to control both the actuator **259** and the downforce control system **214**. Sensor **10000** is not separately illustrated, but it is a reference to any of the following sensors: downforce sensor **238**, force sensor **261**, force sensor **1298**, force sensor **9258**, force sensor **9257**, force sensor **9254**, force sensor **9262**, position sensor **900**, angle sensor **3006**, trench closing sensor **1000**, or load sensor **1610**. Because of the close proximity of the closing system **250** (or its alternatives) to the opening system **220**, the soil is going to be approximately the same in terms of one or more of hardness, moisture, texture, etc. Setting the force for one system (i.e., the closing system **250** or the opening system **220**) will dictate a proportional amount of force needed for the other system (i.e., the opening system **220** or the closing system **250**). For example, if the downforce sensor **238** measures a change in force indicating that the soil is harder and sends a signal to the downforce control system **214** to increase the downforce to the row unit **200**, the same signal can be sent to the actuator **259** to also increase the downforce applied by the actuator **259**. The amount of force can be the same or a portion thereof. The force applied to the downforce control system **214** and the actuator **259** can be proportional to each other. By way of example, the absolute amount of force needed to open a 2 inch deep trench in the soil is greater than the amount of force needed to open a 1 inch deep trench in the soil, but the amount of force required to close the 2 inch deep trench as compared to the 1 inch deep trench may not be the same percentage change as the force needed to open the respective trenches. The system can thus adjust one system relative to the other system for a given depth.

(219) In another embodiment, instead of or in conjunction with any of the various sensors (i.e., force sensor **261**, force sensor **1298**, force sensor **9258**, force sensor **9257**, force sensor **9254**, force sensor **9262**, position sensor **900**, angle sensor **3006**, trench closing sensor **1000** or load sensor **1610**), the force applied by the actuator **259** can be obtained from a downforce prescription map. An example of a downforce prescription map is described in U.S. Patent Application Publication No. 2012/0186503, which is incorporated herein in its entirety by reference. The amount of downforce applied can be prescribed based on one or more of hardness, texture, moisture, organic matter, or any other soil data layer that impacts the force needed to open or close a seed trench.

(220) The following are non-limiting Examples.

(221) Example 1: A wheel comprising: a central axis; a circumferential tread coaxial with the central axis, the circumferential tread extending axially between an inner edge and an outer edge defining an axial width of the wheel; a wall member joined with the circumferential tread, the wall member being continuous between the circumferential tread and a central opening, the wall member extending axially outward from the inner edge of the circumferential tread toward an outer edge of the circumferential tread by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width, the circumferential tread and the wall member together defining a shape in the form of an uppercase Greek letter sigma.

(222) Example 2: The wheel of claim Example 1, wherein the axial distance is equal to or less than one half of the axial width.

(223) Example 3: The wheel of any of any preceding Example, wherein the axial distance is between 10% and one half of the axial width.

(224) Example 4: The wheel of any preceding Example, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

(225) Example 5: The wheel of Example 2, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

(226) Example 6: The wheel of Example 3, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

(227) Example 7: The wheel of any preceding Example, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

(228) Example 8: The wheel of Example 4, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

(229) Example 9: The wheel of Example 5, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

(230) Example 10: The wheel of Example 6, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

(231) Example 11: A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from said main frame; a pair of closing wheels each rotatably supported by said frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, said pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; an actuator supported between said main frame and said frame member such that said actuator is adapted to apply a down force on said frame member; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, wherein the press wheel is the wheel of any of Examples 1 to 10.

(232) Various embodiments of the invention have been described above for purposes of illustrating the details thereof and to enable one of ordinary skill in the art to make and use the invention. The details and features of the disclosed embodiments are not intended to be limiting, as many variations and modifications will be readily apparent to those of skill in the art.

Claims

1. A unitary molded wheel adapted to be rotatably mounted to a mounting arm for packing soil over a seed trench, the unitary molded wheel comprising: a central axis; a circumferential tread coaxial with the central axis, the circumferential tread defining an axial width of the wheel; a wall member unitarily molded with the circumferential tread, the wall member extending radially inward from the circumferential tread to a central opening coaxial with the central axis, the central opening adapted to receive a hub by which the unitary molded wheel is rotatably mounted to the mounting arm, wherein an outer side edge of the unitary molded wheel is oriented away from the mounting arm and an inner side edge of the unitary molded wheel is oriented toward the mounting arm, the wall member being continuous between the circumferential tread and the central opening, the wall member extending axially outward from the inner side edge of the unitary molded wheel toward the outer side edge of the unitary molded wheel by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width; whereby, with the unitary molded wheel mounted to the hub, the continuous wall member prevents soil passing through the unitary molded wheel as the unitary molded wheel rotates over the seed trench packing the soil.
2. The unitary molded wheel of claim 1, wherein the axial distance is equal to or less than one half of the axial width.
3. The unitary molded wheel of claim 1, wherein the axial distance is between 10% and one half of the axial width.
4. The unitary molded wheel of claim 1, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.
5. The unitary molded wheel of claim 2, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.
6. The unitary molded wheel of claim 3, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.
7. The unitary molded wheel of claim 1, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.
8. The unitary molded wheel of claim 4, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.
9. The unitary molded wheel of claim 5, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.
10. The unitary molded wheel of claim 6, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.
11. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame relative to the forward direction of travel; a frame member pivotally supported from the main frame; a first closing wheel rotatably supported by the frame member on a first side of the opened seed trench; a second closing wheel rotatably supported by the frame member on a second side of the opened seed trench, the first and second closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; an actuator supported between the main frame and the frame member such that the actuator is adapted to apply a down force on the frame member; a mounting arm pivotally mounted to the frame member, the mounting arm supporting a unitary molded press wheel rearward of the first and second closing wheels, the unitary molded press wheel comprising: a central axis; a circumferential tread coaxial with the central axis, the circumferential tread defining an axial width of the press wheel; a wall member unitarily molded with the circumferential tread, the wall member extending radially inward from the circumferential tread to a central opening coaxial with the central axis, the central opening adapted to receive a hub by which the unitary molded press wheel is rotatably mounted to the mounting arm, wherein an outer side edge of the unitary molded press wheel is oriented away from the mounting arm and an inner side edge of the unitary molded press wheel is oriented toward the mounting arm, the wall member being continuous between the circumferential tread and the central opening, the wall member extending axially outward from the inner side edge of the unitary molded press wheel toward the outer side edge of the unitary molded press wheel by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width; whereby, with the unitary molded press wheel mounted to the hub, the continuous wall member prevents soil passing through the unitary molded press wheel as the unitary molded press wheel rotates over the seed trench packing the soil.
12. The trench closing system of claim 11, wherein the axial distance is equal to or less than one half of the axial width.
13. The trench closing system of claim 11, wherein the axial distance is between 10% and one half of the axial width.
14. The trench closing system of claim 11, wherein the unitary molded press wheel further includes a plurality of ribs disposed about the

circumferential tread, the ribs extending axially in a direction parallel with the central axis.

15. The trench closing system of claim 12, wherein the unitary molded press wheel further includes a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

16. The trench closing system of claim 13, wherein the unitary molded press wheel further includes a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

17. The trench closing system of claim 11, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

18. The trench closing system of claim 14, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

19. The trench closing system of claim 15, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

20. The trench closing system of claim 16, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.
